1.39 Trigonometry: Introduction

Worked Solutions

1 a)

$A = 40°$.

$\sin 50° = \frac{19}{AC} \Rightarrow AC = \frac{19}{\sin 50°} = 24.8$ cm to 3sf.

$BC = \sqrt{24.802\ldots^2 - 19^2} = 15.9$ cm to 3sf.

1 b)

$AC = \sqrt{30^2 - 26^2} = 15.0$ cm to 3sf.

$\cos B = \frac{26}{30} \Rightarrow B = \arccos \frac{13}{15} = 29.9°$ to 3sf.

$C = 60.1°$ to 3sf.

2)

Let X be the intersection of AC and DB. As $ABCD$ is a kite, the diagonals AC and DB meet at right-angles.

$$\cos 60° = \frac{XA}{1} \Rightarrow XA = \cos 60° = \frac{1}{2}$$
$$\sin 60° = \frac{XD}{1} \Rightarrow XD = \sin 60° = \frac{\sqrt{3}}{2}$$
$$\tan 30° = \frac{XD}{XC} \Rightarrow \tan 30° = \frac{\frac{\sqrt{3}}{2}}{XC}$$
$$\Rightarrow XC = \frac{\frac{\sqrt{3}}{2}}{\tan 30°} = \frac{\frac{\sqrt{3}}{2}}{\frac{\sqrt{3}}{3}} = \frac{3}{2}$$

$$\therefore \text{Area} = 2 \times \frac{1}{2} \times XD \times XA + 2 \times \frac{1}{2} \times XD \times XC$$
$$= XD \times XA + XD \times XC = \frac{\sqrt{3}}{2} \times \frac{1}{2} + \frac{\sqrt{3}}{2} \times \frac{3}{2} = \sqrt{3}\ \text{m}^2$$

3)

$$\tan\theta = \frac{CD}{DE} \Rightarrow \tan\theta = \frac{CD}{1} \Rightarrow CD = \tan\theta$$
$$\text{Area of triangle } CDE = \frac{1}{2} \times 1 \times \tan\theta = \frac{1}{2}\tan\theta$$
$$\text{Area of triangle } ABC = \frac{1}{3} \times \frac{1}{2}\tan\theta = \frac{1}{6}\tan\theta$$
$$\tan\theta = \frac{BC}{AB} \Rightarrow BC = AB\tan\theta$$
$$\frac{1}{2} \times AB \times BC = \frac{1}{6}\tan\theta$$
$$\Rightarrow \frac{1}{2} \times AB \times (AB\tan\theta) = \frac{1}{6}\tan\theta$$
$$\Rightarrow \frac{1}{2}AB^2 = \frac{1}{6} \Rightarrow AB^2 = \frac{1}{3} \Rightarrow AB = \frac{\sqrt{3}}{3}$$

4)

$$
\begin{aligned}
OB^2 &= 1^2 + OA^2 \\
OC^2 &= 1^2 + OB^2 \\
OD^2 &= 1^2 + OC^2 \\
\therefore OD^2 &= 1 + (1 + OB^2) = 1 + (1 + (1 + OA^2)) \\
\Rightarrow OD^2 &= 3 + OA^2 \\
\frac{OD}{OA} &= \frac{\sqrt{3 + OA^2}}{OA} = \frac{\sqrt{3 + OA^2}}{\sqrt{OA^2}} = \sqrt{\frac{3 + OA^2}{OA^2}} = \sqrt{\frac{3}{OA^2} + 1} \\
\sqrt{\frac{3}{OA^2} + 1} = \sqrt{2} &\Rightarrow \frac{3}{OA^2} + 1 = 2 \Rightarrow \frac{3}{OA^2} = 1 \\
\Rightarrow OA^2 &= 3 \Rightarrow OA = \sqrt{3}
\end{aligned}
$$

5)

Area $= \frac{1}{2} \times 23 \times 44 \times \sin 75^\circ = 489$ m^2 to 3sf.

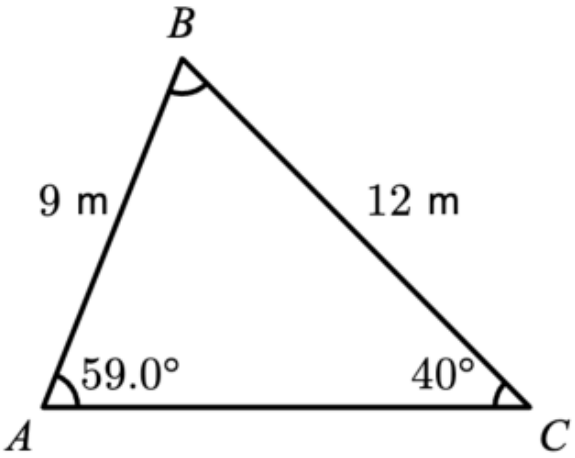
6)

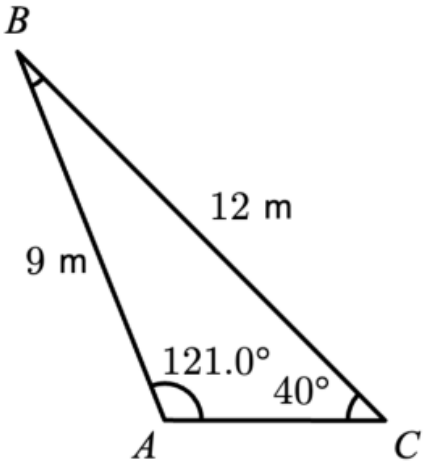
$$
\begin{aligned}
a^2 &= b^2 + c^2 - 2bc\cos A \\
AC^2 &= AB^2 + BC^2 - 2 \times AB \times BC \times \cos(\angle ABC) \\
\Rightarrow AC^2 &= 20^2 + 15^2 - 2 \times 20 \times 15 \times \cos 45^\circ \\
\Rightarrow AC^2 &= 625 - 300\sqrt{2} \\
\Rightarrow AC &= \sqrt{625 - 300\sqrt{2}} = 14.2 \text{ cm to 3sf}
\end{aligned}
$$

7 a)

$$
\begin{aligned}
\frac{\sin 40^\circ}{9} = \frac{\sin A}{12} &\Rightarrow \sin A = \frac{12 \sin 40^\circ}{9} = 0.857\ldots \\
&\Rightarrow A = \arcsin 0.857\ldots = 59.0^\circ \\
&\text{Or } A = 180 - \arcsin 0.857\ldots = 121.0^\circ
\end{aligned}
$$

7 b)





8 a)

$$\frac{\sin A}{8} = \frac{\sin 75^\circ}{11} \Rightarrow \sin A = \frac{8 \sin 75^\circ}{11} = 0.702\ldots$$
$$\Rightarrow A = \arcsin 0.702\ldots = 44.6 \text{ to 3sf}$$
$$\Rightarrow C = 180^\circ - 75^\circ - 44.6\ldots = 60.4^\circ \text{ to 3sf}$$
$$\frac{AB}{\sin C} = \frac{BC}{\sin A} \Rightarrow \frac{AB}{\sin 60.372\ldots} = \frac{8}{\sin 44.627\ldots}$$
$$\Rightarrow AB = \frac{\sin 60.372\ldots}{\sin 44.627\ldots} = 9.90 \text{ to 3sf}$$

8 b)

$$12^2 = 11^2 + 9^2 - 2 \times 11 \times 9 \times \cos A$$
$$\Rightarrow \cos A = \frac{11^2 + 9^2 - 12^2}{2 \times 11 \times 9} = \frac{29}{99}$$
$$\Rightarrow A = \arccos \frac{29}{99} = 73.0^\circ \text{ to 3sf}$$

$$\frac{\sin B}{11} = \frac{\sin 72.9\ldots^\circ}{12} \Rightarrow \sin B = 0.876\ldots$$
$$\Rightarrow B = \arcsin 0.876\ldots = 61.2^\circ \text{ to 3sf}$$
$$C = 180^\circ - 72.9\ldots^\circ - 61.2\ldots^\circ = 45.8^\circ \text{ to 3sf}$$

8 c)

Triangle 1 (this is not how it is drawn in the question, but satisfies the given information):

$$\frac{\sin B}{14} = \frac{\sin 33^\circ}{9} \Rightarrow \sin B = \frac{14 \sin 33^\circ}{9} = 0.847\ldots$$
$$\Rightarrow B = \arcsin 0.847\ldots = 57.9^\circ \text{ to 3sf}$$
$$\Rightarrow A = 180^\circ - 33^\circ - 57.9\ldots = 89.1^\circ \text{ to 3sf}$$
$$BC^2 = 9^2 + 14^2 - 2 \times 9 \times 14 \times \cos 89.0\ldots^\circ$$
$$\Rightarrow BC^2 = 272.997\ldots \Rightarrow BC = 16.5 \text{ m to 3sf}$$

Or

Triangle 2 (this is how it appears to be drawn in the question, with B obtuse and A acute):

$$\frac{\sin B}{14} = \frac{\sin 33^\circ}{9} \Rightarrow \sin B = \frac{14 \sin 33^\circ}{9} = 0.847\ldots$$
$$\Rightarrow B = 180^\circ - \arcsin 0.847\ldots = 122^\circ \text{ to 3sf}$$
$$\Rightarrow A = 180^\circ - 33^\circ - 122.0\ldots = 24.9^\circ \text{ to 3sf}$$
$$BC^2 = 9^2 + 14^2 - 2 \times 9 \times 14 \times \cos 24.9\ldots^\circ$$
$$\Rightarrow BC^2 = 48.443\ldots \Rightarrow BC = 6.96 \text{ m to 3sf}$$

9)

$\angle ACB = 180^\circ - 68^\circ - 58^\circ = 54^\circ$.

$\frac{AC}{\sin 68^\circ} = \frac{3.8}{\sin 54^\circ} \Rightarrow AC = 4.355\ldots$ cm.

Area $\frac{1}{2} \times 3.8 \times 4.355\ldots \times \sin 58^\circ = 7.02$ cm^2 to 3sf.

10)

$25 = \frac{1}{2} \times 16 \times 9 \times \sin x \Rightarrow \sin x = \frac{25}{72}$.

$\Rightarrow x = \arcsin \frac{25}{72} = 20.317\ldots^\circ$.

So $x = 20.3^\circ$ or $180^\circ - 20.317\ldots^\circ = 159.7^\circ$ to 1dp.

11)

$\frac{\sin 30^\circ}{6} = \frac{\sin C}{10} \Rightarrow \sin C = \frac{5}{6} \Rightarrow C = \arcsin \frac{5}{6} = 56.442\ldots^\circ$.

However, C is obtuse, so $C = 180^\circ - 56.442\ldots^\circ = 123.557\ldots^\circ$.

$\angle ABC = 180^\circ - 30^\circ - 123.557\ldots^\circ = 26.442\ldots^\circ$.

Area of triangle $ABC = \frac{1}{2} \times 10 \times 6 \times \sin 26.442\ldots^\circ = 13.359\ldots$ m^2.

Area of triangle $XBC = \frac{1}{2} \times 13.359\ldots = 6.679\ldots$ m^2.

$\therefore \frac{1}{2} \times 6 \times XC = 6.679\ldots \Rightarrow XC = 2.67$ m to 3sf.

12)

$$640 = \frac{1}{2} \times 5m \times 2m^2 \times \sin 45^\circ \Rightarrow 640 = 5m^3 \times \frac{\sqrt{2}}{2}$$
$$\Rightarrow m^3 = 128\sqrt{2} \Rightarrow m^3 = \sqrt{32768}$$
$$\Rightarrow m = \left((32768)^{\frac{1}{2}}\right)^{\frac{1}{3}}$$
$$\Rightarrow m = \left((32768)^{\frac{1}{3}}\right)^{\frac{1}{2}} = 32^{\frac{1}{2}} = 4\sqrt{2}$$
$$\therefore k = 4$$

13 a)

$$(3x+1)^2 = (2x+1)^2 + (3\sqrt{3}x)^2 - 2(2x+1)(3\sqrt{3}x)\cos 30^\circ$$
$$\Rightarrow (3x+1)^2 = (2x+1)^2 + 27x^2 - 2(2x+1)(3\sqrt{3}x) \times \frac{\sqrt{3}}{2}$$
$$\Rightarrow 9x^2 + 6x + 1 = 4x^2 + 4x + 1 + 27x^2 - (2x+1) \times 9x$$
$$\Rightarrow 9x^2 + 6x + 1 = 4x^2 + 4x + 1 + 27x^2 - 18x^2 - 9x$$
$$\Rightarrow 9x^2 + 6x + 1 = 13x^2 - 5x + 1 \Rightarrow 11x - 4x^2 = 0 \text{ as required}$$

13 b)

$11x - 4x^2 = 0 \Rightarrow (11 - 4x)x = 0 \Rightarrow x = \frac{11}{4}$ as $x \neq 0$.

13 c)

The largest angle is BAC.

$$(3\sqrt{3} \times \tfrac{11}{4})^2 = (3 \times \tfrac{11}{4} + 1)^2 + (2 \times \tfrac{11}{4} + 1)^2 - 2 \times (3 \times \tfrac{11}{4} + 1) \times (2 \times \tfrac{11}{4} + 1) \times \cos A$$
$$\Rightarrow \frac{3267}{16} = \frac{1369}{16} + \frac{169}{4} - \frac{481}{4}\cos A$$
$$\Rightarrow \cos A = \frac{\frac{1369}{16} + \frac{169}{4} - \frac{3267}{16}}{\frac{481}{4}} = -\frac{47}{74}$$
$$\Rightarrow A = \arccos(-\tfrac{47}{74}) = 129^\circ \text{ to 3sf}$$

14)

$$\text{Area} = 2 \times \frac{1}{2} \times 0.86 \times 1.4 \times \sin 120^\circ = 1.204 \times \frac{\sqrt{3}}{2}$$
$$= 0.602 \times \sqrt{3} = \frac{301}{500}\sqrt{3} \text{ m}^2$$

15)

As x is the largest angle, it must be opposite the largest side, which is $b+2$.

$$\begin{aligned}\therefore (b+2)^2 &= (b+1)^2 + b^2 - 2(b+1)b\cos x \\ \Rightarrow \cos x &= \frac{(b+1)^2 + b^2 - (b+2)^2}{2(b+1)b} = \frac{b^2+2b+1+b^2-(b^2+4b+4)}{2(b+1)b} \\ &= \frac{b^2-2b-3}{2(b+1)b} = \frac{(b+1)(b-3)}{2(b+1)b} = \frac{b-3}{2b}\end{aligned}$$

16)

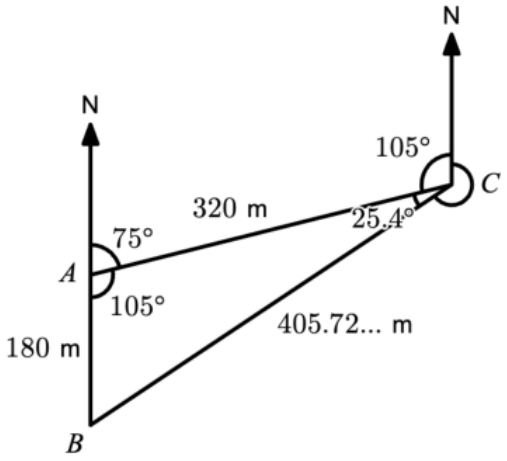
$\angle ABC = 180° - 42.7° = 137.3°$.

$\angle BCA = 180° - 137.3° - 30.5° = 12.2°$.

$\frac{BC}{\sin 30.5°} = \frac{20}{\sin 12.2°} \Rightarrow BC = \frac{20 \sin 30.5°}{\sin 12.2°} = 48.033\ldots$ m.

$\sin 42.7° = \frac{OC}{48.033...} \Rightarrow OC = 32.6$ m to 3sf.

17)



17 a)

Bearing 180°.

17 b)

$360° - 105° = 255°$.

Bearing 255°.

17 c)

$$\begin{aligned}\angle BAC &= 180° - 75 = 105° \\ BC^2 &= AB^2 + AC^2 - 2 \times AB \times AC \times \cos 105° \\ &= 180^2 + 320^2 - 2 \times 180 \times 320 \times \cos 105° \\ &= 164615.95\ldots \Rightarrow BC = 405.7289\ldots \text{ m}\end{aligned}$$

$$\frac{\sin(ACB)}{180} = \frac{\sin 105^\circ}{405.7289\ldots}$$
$$\Rightarrow \angle ACB = \arcsin\left(\frac{180\sin 105^\circ}{405.7289\ldots}\right) = 25.374\ldots^\circ$$
$$360^\circ - 105^\circ - 25.374\ldots^\circ = 229.625\ldots^\circ$$

Bearing 230°.

18)

$$10^2 = p^2 + (p+5)^2 - 2 \times p \times (p+5) \times \cos 60^\circ$$
$$\Rightarrow 100 = p^2 + (p+5)^2 - p(p+5)$$
$$\Rightarrow 100 = p^2 + p^2 + 10p + 25 - p^2 - 5p$$
$$\Rightarrow 0 = p^2 + 5p - 75 \Rightarrow p = \frac{-5 \pm 5\sqrt{13}}{2}$$

$p > 0$ so $p = \frac{-5+5\sqrt{13}}{2}$.

19)

Divide the pentagon into five triangles, so each triangle will have an area of $\frac{60}{5} = 12$ cm^2.

$\frac{360}{5} = 72^\circ$.

We will call the distance from the centre of the pentagon to a vertex r.

$\frac{1}{2} \times r^2 \times \sin 72^\circ = 12 \Rightarrow r = 5.023\ldots$ cm.

$l^2 = 5.023\ldots^2 + 5.023\ldots^2 - 2 \times 5.023\ldots \times 5.023\ldots \times \cos 72^\circ \Rightarrow l = 5.91$ cm to 3sf.

20)

Divide the octagon into eight triangles, so each triangle will have an area of $\frac{450}{8} = 56.25$ cm^2.

$\frac{360}{8} = 45^\circ$.

We will call the distance from the centre of the octagon to a vertex r.

$\frac{1}{2} \times r^2 \times \sin 45^\circ = 56.25 \Rightarrow r = 12.61\ldots$ cm.

$l^2 = 12.61\ldots^2 + 12.61\ldots^2 - 2 \times 12.61\ldots \times 12.61\ldots \times \cos 45^\circ \Rightarrow l = 9.65$ cm to 3sf.

21 a)

Triangle XBC has area 10 units^2, so shortest distance from X to BC must be 2 as $\frac{1}{2} \times 2 \times 10 = 10$. Likewise, triangle XAB has area 20 units^2, so shortest distance from X to AB must be 4 as $\frac{1}{2} \times 4 \times 10 = 20$.

$\therefore X$ has coordinates $(8, 6)$, so $p = 8$ and $q = 6$.

21 b)

Area of triangle XAD is 40 units^2.

So area of triangle XYD is at least 20 units^2.

$XD = \sqrt{8^2 + 6^2} = 10$.

YX and DC are parallel, so $\angle YXD = \angle XDC = \arctan\frac{6}{8} = \arctan\frac{3}{4} = 36.869\ldots^\circ$.

$\therefore \frac{1}{2} \times XY \times 10 \times \sin(\arctan\frac{3}{4}) \geq 20 \Rightarrow XY \geq \frac{20}{3}$.

$\therefore 0 < r \leq (8 - \frac{20}{3}) = \frac{4}{3}$.
$\Rightarrow 0 < r \leq \frac{4}{3}$.

